

Trabajo Práctico N° 1: **Vectores y Espacios Vectoriales.**

Ejercicio 1.

Dados los vectores $u = (1, 2)$, $v = (-1, 3)$ y $w = (-1, -2)$, calcular analítica y gráficamente las siguientes operaciones.

(a)

$$u + v = (1, 2) + (-1, 3)$$

$$u + v = (0, 5).$$

Gráfico.

(b)

$$v + w = (-1, 3) + (-1, -2)$$

$$v + w = (-2, 1).$$

Gráfico.

(c)

$$3u + 3v = 3(1, 2) + 3(-1, 3)$$

$$3u + 3v = (3, 6) + (-3, 9)$$

$$3u + 3v = (0, 15).$$

Gráfico.

(d)

$$3(u + v) = 3[(1, 2) + (-1, 3)]$$

$$3(u + v) = 3(0, 5)$$

$$3(u + v) = (0, 15).$$

Gráfico.

(e)

$$(u + v) + w = [(1, 2) + (-1, 3)] + (-1, 2)$$

$$(u + v) + w = (0, 5) + (-1, 2)$$

$$(u + v) + w = (-1, 7).$$

Gráfico.

(f)

$$u + (v + w) = (1, 2) + [(-1, 3) + (-1, -2)]$$

$$u + (v + w) = (1, 2) + (-2, 1)$$

$$u + (v + w) = (-1, 3).$$

Gráfico.

(g)

$$u - v = (1, 2) - (-1, 3)$$

$$u - v = (4, -1).$$

Gráfico.

(h)

$$u + (v - w) = (1, 2) + [(-1, 3) - (-1, -2)]$$

$$u + (v - w) = (1, 2) + (0, 5)$$

$$u + (v - w) = (1, 7).$$

Gráfico.

(i)

$$\frac{5}{4}u + \frac{1}{2}v - \frac{3}{2}w = \frac{5}{4}(1, 2) + \frac{1}{2}(-1, 3) - \frac{3}{2}(-1, -2)$$

$$\frac{5}{4}u + \frac{1}{2}v - \frac{3}{2}w = \left(\frac{5}{4}, \frac{5}{2}\right) + \left(\frac{-1}{2}, \frac{3}{2}\right) + \left(\frac{3}{2}, 3\right)$$

$$\frac{5}{4}u + \frac{1}{2}v - \frac{3}{2}w = \left(\frac{9}{4}, 7\right).$$

Gráfico.

Ejercicio 2.

Sea $w = (1, 3) \in \mathbb{R}^2$ un vector. Graficar en el plano.

(a) $L = \{tw : t \in \mathbb{R}\}.$

Gráfico.

(b) $L = \{tw : t \in \mathbb{R}_{\geq 0}\}.$

Gráfico.

(c) $L = \{tw : t \in \mathbb{R}, 0 \leq t \leq 1\}.$

Gráfico.

Ejercicio 3.

Dados los vectores $u = (0, 1, 2)$, $v = (1, 1, 0)$ y $w = (-1, 1, 1)$, calcular las operaciones:

(a)

$$\begin{aligned}u + v &= (0, 1, 2) + (1, 1, 0) \\u + v &= (1, 2, 2).\end{aligned}$$

(b)

$$\begin{aligned}u + v + w &= (0, 1, 2) + (1, 1, 0) + (-1, 1, 1) \\u + v + w &= (0, 3, 3).\end{aligned}$$

(c)

$$\begin{aligned}u - v &= (0, 1, 2) - (1, 1, 0) \\u - v &= (-1, 0, 2).\end{aligned}$$

(d)

$$\begin{aligned}2u &= 2(0, 1, 2) \\2u &= (0, 2, 4).\end{aligned}$$

(e)

$$\begin{aligned}-3v &= -3(1, 1, 0) \\-3v &= (-3, -3, 0).\end{aligned}$$

(f)

$$\begin{aligned}-v + \frac{2}{3}w &= -(1, 1, 0) + \frac{2}{3}(-1, 1, 1) \\-v + \frac{2}{3}w &= (-1, -1, 0) + \left(\frac{-2}{3}, \frac{2}{3}, \frac{2}{3}\right) \\-v + \frac{2}{3}w &= \left(\frac{-5}{3}, \frac{-1}{3}, \frac{2}{3}\right).\end{aligned}$$

Ejercicio 4.

Hallar x e y para que los vectores v y w resulten iguales.

(a) $v = (x, 3)$ y $w = (2, x + y)$.

$$v = w$$

$$(x, 3) = (2, x + y).$$

$$\begin{cases} x = 2 \\ 3 = x + y \end{cases}$$

$$x + y = 3$$

$$y = -x + 3$$

$$y = -2 + 3$$

$$y = 1.$$

Por lo tanto, los valores de x e y para que los vectores v y w resulten iguales son $(2, 1)$.

(b) $v = (4, y)$ y $w = x(2, 3)$.

$$v = w$$

$$(4, y) = x(2, 3)$$

$$(4, y) = (2x, 3x).$$

$$\begin{cases} 4 = 2x \\ y = 3x \end{cases}$$

$$x = \frac{4}{2}$$

$$x = 2.$$

$$y = 3 \cdot 2$$

$$y = 6.$$

Por lo tanto, los valores de x e y para que los vectores v y w resulten iguales son $(2, 6)$.

(c) $v = x(3, 2)$ y $w = 2(y, -1)$.

$$v = w$$

$$x(3, 2) = 2(y, -1)$$

$$(3x, 2x) = (2y, -2).$$

$$\begin{cases} 3x = 2y \\ 2x = -2 \end{cases}$$

$$x = \frac{-2}{2}$$

$$x = -1.$$

$$2y = 3x$$

$$y = \frac{3}{2}x$$

$$y = \frac{3}{2}(-1)$$

$$y = \frac{-3}{2}.$$

Por lo tanto, los valores de x e y para que los vectores v y w resulten iguales son $(-1, \frac{-3}{2})$.

(d) $v = x(2, y)$ y $w = y(1, -2)$.

$$v = w$$

$$x(2, y) = y(1, -2)$$

$$(2x, xy) = (y, -2y).$$

$$\begin{cases} 2x = y \\ xy = -2y \end{cases}$$

$$x2x = -2 * 2x$$

$$2x^2 = -4x$$

$$x^2 = \frac{-4}{2}x$$

$$x^2 = -2x$$

$$x^2 + 2x = 0$$

$$x(x + 2) = 0.$$

$$x_1 = 0.$$

$$x_2 = -2.$$

$$y_1 = 2 * 0 = 0.$$

$$y_2 = 2(-2) = -4.$$

Por lo tanto, los valores de x e y para que los vectores v y w resulten iguales son $(0, 0)$ y $(-2, -4)$.

Ejercicio 5

Normalizar los siguientes vectores.

(a) $u = (-3, 1, -2, 4, -5)$.

$$\begin{aligned} u' &= \frac{u}{\|u\|} \\ u' &= \frac{1}{\sqrt{(-3)^2 + 1^2 + (-2)^2 + 4^2 + (-5)^2}} (-3, 1, -2, 4, -5) \\ u' &= \frac{1}{\sqrt{9+1+4+16+25}} (-3, 1, -2, 4, -5) \\ u' &= \frac{1}{\sqrt{55}} (-3, 1, -2, 4, -5) \\ u' &= \left(\frac{-3}{\sqrt{55}}, \frac{1}{\sqrt{55}}, \frac{-2}{\sqrt{55}}, \frac{4}{\sqrt{55}}, \frac{-5}{\sqrt{55}} \right) \\ u' &= \left(\frac{-3\sqrt{55}}{55}, \frac{\sqrt{55}}{55}, \frac{-2\sqrt{55}}{55}, \frac{4\sqrt{55}}{55}, \frac{-5\sqrt{55}}{55} \right) \\ u' &= \left(\frac{-3\sqrt{55}}{55}, \frac{\sqrt{55}}{55}, \frac{-2\sqrt{55}}{55}, \frac{4\sqrt{55}}{55}, \frac{-\sqrt{55}}{11} \right). \end{aligned}$$

(b) $v = (4, -2, -3, 8)$.

$$\begin{aligned} v' &= \frac{v}{\|v\|} \\ v' &= \frac{1}{\sqrt{4^2 + (-2)^2 + (-3)^2 + 8^2}} (4, -2, -3, 8) \\ v' &= \frac{1}{\sqrt{16+4+9+64}} (4, -2, -3, 8) \\ v' &= \frac{1}{\sqrt{93}} (4, -2, -3, 8) \\ v' &= \left(\frac{4}{\sqrt{93}}, \frac{-2}{\sqrt{93}}, \frac{-3}{\sqrt{93}}, \frac{8}{\sqrt{93}} \right) \\ v' &= \left(\frac{4\sqrt{93}}{93}, \frac{-2\sqrt{93}}{93}, \frac{-3\sqrt{93}}{93}, \frac{8\sqrt{93}}{93} \right) \\ v' &= \left(\frac{4\sqrt{93}}{93}, \frac{-2\sqrt{93}}{93}, \frac{-\sqrt{93}}{31}, \frac{8\sqrt{93}}{93} \right). \end{aligned}$$

(c) $w = \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{4} \right)$.

$$\begin{aligned} w' &= \frac{w}{\|w\|} \\ w' &= \frac{1}{\sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{2}{3}\right)^2 + \left(\frac{-1}{4}\right)^2}} \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{4} \right) \\ w' &= \frac{1}{\sqrt{\frac{1}{4} + \frac{4}{9} + \frac{1}{16}}} \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{4} \right) \\ w' &= \frac{1}{\sqrt{\frac{109}{144}}} \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{4} \right) \\ w' &= \frac{1}{\frac{\sqrt{109}}{12}} \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{4} \right) \end{aligned}$$

$$\begin{aligned}
 w' &= \frac{12}{\sqrt{109}} \left(\frac{1}{2}, \frac{2}{3}, \frac{-1}{4} \right) \\
 w' &= \left(\frac{6}{\sqrt{109}}, \frac{8}{\sqrt{109}}, \frac{-3}{\sqrt{109}} \right) \\
 w' &= \left(\frac{6\sqrt{109}}{109}, \frac{8\sqrt{109}}{109}, \frac{-3\sqrt{109}}{4 \cdot 109} \right) \\
 w' &= \left(\frac{6\sqrt{109}}{109}, \frac{8\sqrt{109}}{109}, \frac{-3\sqrt{109}}{109} \right).
 \end{aligned}$$

Ejercicio 6.

Determinar todos los valores de $k \in \mathbb{R}$ que verifican:

(a) El vector $u = (4, k)$ tiene norma 5.

$$\begin{aligned} \|u\| &= 5 \\ \sqrt{4^2 + k^2} &= 5 \\ \sqrt{16 + k^2} &= 5 \\ 16 + k^2 &= 5^2 \\ 16 + k^2 &= 25 \\ k^2 &= 25 - 16 \\ k^2 &= 9 \\ \sqrt{k^2} &= \sqrt{9} \\ |k| &= 3 \\ k &= \pm 3. \end{aligned}$$

(b) El vector $v = (1, k, 0)$ tiene norma 2.

$$\begin{aligned} \|v\| &= 2 \\ \sqrt{1^2 + k^2 + 0^2} &= 2 \\ \sqrt{1 + k^2 + 0} &= 2 \\ \sqrt{1 + k^2} &= 2 \\ 1 + k^2 &= 2^2 \\ 1 + k^2 &= 4 \\ k^2 &= 4 - 1 \\ k^2 &= 3 \\ \sqrt{k^2} &= \sqrt{3} \\ |k| &= \sqrt{3} \\ k &= \pm \sqrt{3}. \end{aligned}$$

(c) El vector $w = k(2, 2, 1)$ tiene norma 1.

$$\begin{aligned} \|w\| &= 1 \\ \sqrt{(2k)^2 + (2k)^2 + 1^2} &= 1 \\ \sqrt{4k^2 + 4k^2 + 1} &= 1 \\ \sqrt{8k^2 + 1} &= 1 \\ 8k^2 + 1 &= 1^2 \\ 8k^2 + 1 &= 1 \\ 8k^2 &= 1 - 1 \\ 8k^2 &= 0 \\ k^2 &= \frac{0}{8} \end{aligned}$$

$$k^2 = 0$$

$$\sqrt{k^2} = \sqrt{0}$$

$$|k| = 0$$

$$k = \pm 0$$

$$k = 0.$$

(d) El vector $z = (1, k, -2, 5)$ tiene norma $\sqrt{39}$.

$$\|z\| = \sqrt{39}$$

$$\sqrt{1^2 + k^2 + (-2)^2 + 5^2} = \sqrt{39}$$

$$\sqrt{1 + k^2 + 4 + 25} = \sqrt{39}$$

$$\sqrt{30 + k^2} = \sqrt{39}$$

$$30 + k^2 = (\sqrt{39})^2$$

$$30 + k^2 = 39$$

$$k^2 = 39 - 30$$

$$k^2 = 9$$

$$\sqrt{k^2} = \sqrt{9}$$

$$|k| = 3$$

$$k = \pm 3.$$

Ejercicio 7.

Dados los vectores $v = (1, -2, 2)$, $w = (2, 0, 3)$ y $z = (4, 4, 4)$, realizar las operaciones.

(a)

$$\begin{aligned} v * w &= (1, -2, 2) * (2, 0, 3) \\ v * w &= 1 * 2 - 2 * 0 + 2 * 3 \\ v * w &= 2 - 0 + 6 \\ v * w &= 8. \end{aligned}$$

(b)

$$\begin{aligned} w * v &= (2, 0, 3) * (1, -2, 2) \\ w * v &= 2 * 1 + 0 * (-2) + 3 * 2 \\ w * v &= 2 + 0 + 6 \\ w * v &= 8. \end{aligned}$$

(c)

$$\begin{aligned} (v + w) * z &= [(1, -2, 2) + (2, 0, 3)] * (4, 4, 4) \\ (v + w) * z &= (3, -2, 5) * (4, 4, 4) \\ (v + w) * z &= 3 * 4 - 2 * 4 + 5 * 4 \\ (v + w) * z &= 12 - 8 + 20 \\ (v + w) * z &= 24. \end{aligned}$$

(d)

$$\begin{aligned} (v * z) + (w * z) &= (1, -2, 2) * (4, 4, 4) + (2, 0, 3) * (4, 4, 4) \\ (v * z) + (w * z) &= [(1, -2, 2) + (2, 0, 3)] * (4, 4, 4) \\ (v * z) + (w * z) &= (3, -2, 5) * (4, 4, 4) \\ (v * z) + (w * z) &= 3 * 4 - 2 * 4 + 5 * 4 \\ (v * z) + (w * z) &= 12 - 8 + 20 \\ (v * z) + (w * z) &= 24. \end{aligned}$$

(e)

$$\begin{aligned} 3v * w &= 3 (1, -2, 2) * (2, 0, 3) \\ 3v * w &= (3, -6, 12) * (2, 0, 3) \\ 3v * w &= 3 * 2 - 6 * 0 + 12 * 3 \\ 3v * w &= 6 - 0 + 36 \end{aligned}$$

$$3v * w = 42.$$

(f)

$$v * 3w = (1, -2, 2) * 3(2, 0, 3)$$

$$v * 3w = (1, -2, 2) * (6, 0, 9)$$

$$v * 3w = 1 * 6 - 2 * 0 + 2 * 9$$

$$v * 3w = 6 - 0 + 18$$

$$v * 3w = 24.$$

(g)

$$3(v * w) = 3[(1, -2, 2) * (2, 0, 3)]$$

$$3(v * w) = 3(1 * 2 - 2 * 0 + 2 * 3)$$

$$3(v * w) = 3(2 - 0 + 6)$$

$$3(v * w) = 3 * 8$$

$$3(v * w) = 24.$$

(h)

$$v * v = (1, -2, 2) * (1, -2, 2)$$

$$v * v = 1 * 1 - 2 * (-2) + 2 * 2$$

$$v * v = 1 - 4 + 4$$

$$v * v = 1.$$

(i)

$$w * w = (2, 0, 3) * (2, 0, 3)$$

$$w * w = 2 * 2 + 0 * 0 + 3 * 3$$

$$w * w = 4 + 0 + 9$$

$$w * w = 13.$$

Ejercicio 8.

En cada uno de los siguientes casos, calcular el ángulo entre los vectores u y v .

(a) $u = (1, 1)$ y $v = (1, -1)$.

$$u \cdot v = (1, 1) \cdot (1, -1)$$

$$u \cdot v = 1 \cdot 1 + 1 \cdot (-1)$$

$$u \cdot v = 1 - 1$$

$$u \cdot v = 0.$$

$$\varphi = \arccos \left(\frac{\cos(\varphi)}{\|u\| \|v\|} \right)$$

$$\varphi = \arccos(0)$$

$$\varphi = 90^\circ.$$

Por lo tanto, u y v son ortogonales.

(b) $u = (3, -1, 2)$ y $v = (4, 3, -1)$.

$$u \cdot v = (3, -1, 2) \cdot (4, 3, -1)$$

$$u \cdot v = 3 \cdot 4 - 1 \cdot 3 + 2 \cdot (-1)$$

$$u \cdot v = 12 - 3 - 2$$

$$u \cdot v = 7.$$

$$\|u\| = \sqrt{3^2 + (-1)^2 + 2^2}$$

$$\|u\| = \sqrt{9 + 1 + 4}$$

$$\|u\| = \sqrt{14}.$$

$$\|v\| = \sqrt{4^2 + 3^2 + (-1)^2}$$

$$\|v\| = \sqrt{16 + 9 + 1}$$

$$\|v\| = \sqrt{26}.$$

$$\varphi = \arccos \left(\frac{u \cdot v}{\|u\| \|v\|} \right)$$

$$\varphi = \arccos \left(\frac{7}{\sqrt{14} \sqrt{26}} \right)$$

$$\varphi = \arccos \left(\frac{7}{\sqrt{14 \cdot 26}} \right)$$

$$\varphi = \arccos \left(\frac{7}{\sqrt{390}} \right)$$

$$\varphi = 69,24^\circ.$$

Por lo tanto, u y v no son ortogonales.

(c) $u = (1, -2, 3)$ y $v = (2, 5, 4)$.

$$\begin{aligned}u \cdot v &= (1, -2, 3) \cdot (2, 5, 4) \\u \cdot v &= 1 \cdot 2 - 2 \cdot 5 + 3 \cdot 4 \\u \cdot v &= 2 - 10 + 12 \\u \cdot v &= 4.\end{aligned}$$

$$\begin{aligned}\|u\| &= \sqrt{1^2 + (-2)^2 + 3^2} \\ \|u\| &= \sqrt{1 + 4 + 9} \\ \|u\| &= \sqrt{15}.\end{aligned}$$

$$\begin{aligned}\|v\| &= \sqrt{2^2 + 5^2 + 4^2} \\ \|v\| &= \sqrt{4 + 25 + 16} \\ \|v\| &= \sqrt{40}.\end{aligned}$$

$$\begin{aligned}\varphi &= \arccos\left(\frac{u \cdot v}{\|u\|\|v\|}\right) \\ \varphi &= \arccos\left(\frac{4}{\sqrt{15}\sqrt{40}}\right) \\ \varphi &= \arccos\left(\frac{4}{\sqrt{15 \cdot 40}}\right) \\ \varphi &= \arccos\left(\frac{4}{\sqrt{600}}\right) \\ \varphi &= 80,6^\circ.\end{aligned}$$

Por lo tanto, u y v no son ortogonales.

Ejercicio 9.

Se consideran los vectores de $\mathbb{R}^3$ $u = (1, -3, 2)$ y $v = (2, -1, 1)$.

(a) Escribir al vector $w = (1, 7, -4)$ como combinación lineal de u y v .

$$w = au + bv$$

$$(1, 7, -4) = a(1, -3, 2) + b(2, -1, 1)$$

$$(1, 7, -4) = (a, -3a, 2a) + (2b, -b, b)$$

$$(1, 7, -4) = (a + 2b, -3a - b, 2a + b).$$

$$\begin{cases} 1 = a + 2b \\ 7 = -3a - b \\ -4 = 2a + b \end{cases}$$

$$2b = 1 - a$$

$$b = \frac{1-a}{2}$$

$$b = \frac{1}{2} - \frac{1}{2}a.$$

$$7 = -3a - \left(\frac{1}{2} - \frac{1}{2}a\right)$$

$$7 = -3a - \frac{1}{2} + \frac{1}{2}a$$

$$7 = \frac{-5}{2}a - \frac{1}{2}$$

$$\frac{5}{2}a = \frac{-1}{2} - 7$$

$$\frac{5}{2}a = \frac{-15}{2}$$

$$a = \frac{-15}{5}$$

$$a = -3.$$

$$b = \frac{1}{2} - \frac{1}{2}(-3)$$

$$b = \frac{1}{2} + \frac{3}{2}$$

$$b = 2.$$

$$-4 = 2(-3) + 2$$

$$-4 = -6 + 2$$

$$-4 = -4.$$

Por lo tanto, se escribe al vector w como combinación lineal de u y v de la siguiente manera:

$$w = -3u + 2v$$

$$(1, 7, -4) = -3(1, -3, 2) + 2(2, -1, 1).$$

(b) Escribir al vector $z = (2, -5, 4)$ como combinación lineal de u y v .

$$z = au + bv$$

$$(2, -5, 4) = a(1, -3, 2) + b(2, -1, 1)$$

$$(2, -5, 4) = (a, -3a, 2a) + (2b, -b, b)$$

$$(2, -5, 4) = (a + 2b, -3a - b, 2a + b).$$

$$\begin{cases} 2 = a + 2b \\ -5 = -3a - b. \\ 4 = 2a + b \end{cases}$$

$$2b = 2 - a$$

$$b = \frac{2-a}{2}$$

$$b = 1 - \frac{1}{2}a.$$

$$-5 = -3a - (1 - \frac{1}{2}a)$$

$$-5 = -3a - 1 + \frac{1}{2}a$$

$$-5 = -\frac{5}{2}a - 1$$

$$\frac{5}{2}a = -1 + 5$$

$$\frac{5}{2}a = 4$$

$$a = \frac{4}{\frac{5}{2}}$$

$$a = \frac{8}{5}.$$

$$b = 1 - \frac{1}{2} \cdot \frac{8}{5}$$

$$b = 1 - \frac{4}{5}$$

$$b = \frac{1}{5}.$$

$$4 = 2 \cdot \frac{8}{5} + \frac{1}{5}$$

$$4 = \frac{16}{5} + \frac{1}{5}$$

$$4 \neq \frac{17}{5}.$$

Por lo tanto, no es posible escribir al vector z como combinación lineal de u y v .

(c) ¿Para qué valores de k el vector $y = (1, k, 5)$ es una combinación lineal de u y v ?

$$y = au + bv$$

$$(1, k, 5) = a(1, -3, 2) + b(2, -1, 1)$$

$$(1, k, 5) = (a, -3a, 2a) + (2b, -b, b)$$

$$(1, k, 5) = (a + 2b, -3a - b, 2a + b).$$

$$\begin{cases} 1 = a + 2b \\ k = -3a - b \\ 5 = 2a + b \end{cases}$$

$$2b = 1 - a$$

$$b = \frac{1-a}{2}$$

$$b = \frac{1}{2} - \frac{1}{2}a.$$

$$5 = 2a + \frac{1}{2} - \frac{1}{2}a$$

$$5 = \frac{3}{2}a + \frac{1}{2}$$

$$\frac{3}{2}a = 5 - \frac{1}{2}$$

$$\frac{3}{2}a = \frac{9}{2}$$

$$a = \frac{\frac{9}{2}}{\frac{3}{2}}$$

$$a = 3.$$

$$b = \frac{1}{2} - \frac{1}{2} * 3$$

$$b = \frac{1}{2} - \frac{3}{2}$$

$$b = -1.$$

$$k = -3 * 3 - (-1)$$

$$k = -9 + 1$$

$$k = -8.$$

Por lo tanto, el vector y es una combinación lineal de u y v para $k = -8$.

Ejercicio 10.

Estudiar si el conjunto de vectores $S = \{(2, 1, 0), (3, 1, 1), (3, 2, -1)\}$ es una base $\mathbb{R}^3$.

$$a(2, 1, 0) + b(3, 1, 1) + c(3, 2, -1) = (0, 0, 0)$$

$$(2a, a, 0) + (3b, b, b) + (3c, 2c, -c) = (0, 0, 0)$$

$$(2a + 3b + 3c, a + b + 2c, b - c) = (0, 0, 0).$$

$$\begin{cases} 2a + 3b + 3c = 0 \\ a + b + 2c = 0 \\ b - c = 0 \end{cases}.$$

$$b = c.$$

$$a + c + 2c = 0$$

$$a + 3c = 0$$

$$a = -3c.$$

$$2(-3c) + 3c + 3c = 0$$

$$-6c + 6c = 0$$

$$0 = 0.$$

Por lo tanto, S no es linealmente independiente y, entonces, no es una base de $\mathbb{R}^3$.

Ejercicio 11.

Encontrar un sistema generador del subespacio de $\mathbb{R}^4$ definido por:

$$T = \{(x, y, z, t) \in \mathbb{R}^4 : x + 2z - 3t + 2y = 0\}.$$

$$x = -2z + 3t - 2y.$$

$$(x, y, z, t) = (-2z + 3t - 2y, y, z, t)$$

$$(x, y, z, t) = y(-2, 1, 0, 0) + z(-2, 0, 1, 0) + t(3, 0, 0, 1).$$

$$T = \langle (-2, 1, 0, 0), (-2, 0, 1, 0), (3, 0, 0, 1) \rangle.$$

Ejercicio 12.

Determinar si los siguientes conjuntos son bases de $\mathbb{R}^3$.

(a) $\{(1, 1, 4), (0, 2, 1), (3, 1, 9)\}$.

$$a(1, 1, 4) + b(0, 2, 1) + c(3, 1, 9) = (0, 0, 0)$$

$$(a, a, 4a) + (0, 2b, b) + (3c, c, 9c) = (0, 0, 0)$$

$$(a + 3c, a + 2b + c, 4a + b + 9c) = (0, 0, 0).$$

$$\begin{cases} a + 3c = 0 \\ a + 2b + c = 0 \\ 4a + b + 9c = 0 \end{cases}$$

$$3c = -a$$

$$c = \frac{-1}{3} a.$$

$$a + 2b - \frac{1}{3}a = 0$$

$$\frac{2}{3}a + 2b = 0$$

$$2b = \frac{-2}{3}a$$

$$b = \frac{\frac{-2}{3}}{2}a$$

$$b = \frac{-1}{3}a.$$

$$4a - \frac{1}{3}a + 9\left(\frac{-1}{3}a\right) = 0$$

$$4a - \frac{1}{3}a - 3a = 0$$

$$\frac{2}{3}a = 0$$

$$a = \frac{0}{\frac{2}{3}}$$

$$a = 0.$$

$$b = \frac{-1}{3} * 0$$

$$b = 0.$$

$$c = \frac{-1}{3} * 0$$

$$c = 0.$$

$$a = b = c = 0.$$

$$a(1, 1, 4) + b(0, 2, 1) + c(3, 1, 9) = (x, y, z)$$

$$(a, a, 4a) + (0, 2b, b) + (3c, c, 9c) = (x, y, z)$$

$$(a + 3c, a + 2b + c, 4a + b + 9c) = (x, y, z).$$

$$\begin{cases} a + 3c = x \\ a + 2b + c = y \\ 4a + b + 9c = z \end{cases}$$

$$3c = x - a$$

$$c = \frac{x-a}{3}$$

$$c = \frac{1}{3}x - \frac{1}{3}a.$$

$$a + 2b + \frac{1}{3}x - \frac{1}{3}a = y$$

$$\frac{2}{3}a + 2b + \frac{1}{3}x = y$$

$$2b = y - \frac{2}{3}a - \frac{1}{3}x$$

$$b = \frac{y - \frac{2}{3}a - \frac{1}{3}x}{2}$$

$$b = \frac{1}{2}y - \frac{1}{3}a - \frac{1}{6}x.$$

$$4a + \frac{1}{2}y - \frac{1}{3}a - \frac{1}{6}x + 9\left(\frac{1}{3}x - \frac{1}{3}a\right) = z$$

$$4a + \frac{1}{2}y - \frac{1}{3}a - \frac{1}{6}x + 3x - 3a = z$$

$$\frac{2}{3}a + \frac{17}{6}x + \frac{1}{2}y = z$$

$$\frac{2}{3}a = \frac{-17}{6}x - \frac{1}{2}y + z$$

$$a = \frac{\frac{-17}{6}x - \frac{1}{2}y + z}{\frac{2}{3}}$$

$$a = \frac{-51}{12}x - \frac{3}{4}y + \frac{3}{2}z$$

$$a = \frac{-51x - 9y + 18z}{12}.$$

$$b = \frac{1}{2}y - \frac{1}{3}\left(\frac{-51}{12}x - \frac{3}{4}y + \frac{3}{2}z\right) - \frac{1}{6}x$$

$$b = \frac{1}{2}y + \frac{17}{12}x + \frac{1}{4}y - \frac{1}{2}z - \frac{1}{6}x$$

$$b = \frac{15}{12}x + \frac{3}{4}y - \frac{1}{2}z$$

$$b = \frac{15x + 9y - 6z}{12}.$$

$$c = \frac{1}{3}x - \frac{1}{3}\left(\frac{-51}{12}x - \frac{3}{4}y + \frac{3}{2}z\right)$$

$$c = \frac{1}{3}x - \frac{17}{12}x + \frac{1}{4}y - \frac{1}{2}z$$

$$c = \frac{-13}{12}x + \frac{1}{4}y - \frac{1}{2}z$$

$$c = \frac{-13x + 3y - 6z}{12}.$$

Por lo tanto, este conjunto es base de $\mathbb{R}^3$.

(b) $\{(1, 0, 1), (1, 1, 0)\}$.

Este conjunto no es base de $\mathbb{R}^3$, ya que $\dim(\mathbb{R}^3) = 3$.

$$(c) \{(2, 1, 1), (2, 2, 1), (2, 2, -1)\}.$$

$$\begin{aligned} a(2, 1, 1) + b(2, 2, 1) + c(2, 2, -1) &= (0, 0, 0) \\ (2a, a, a) + (2b, 2b, b) + (2c, 2c, -c) &= (0, 0, 0) \\ (2a + 2b + 2c, a + 2b + 2c, a + b - c) &= (0, 0, 0). \end{aligned}$$

$$\begin{cases} 2a + 2b + 2c = 0 \\ a + 2b + 2c = 0 \\ a + b - c = 0 \end{cases}.$$

$$\begin{aligned} 2(a + b + c) &= 0 \\ a + b + c &= \frac{0}{2} \\ a + b + c &= 0 \\ c &= -a - b. \end{aligned}$$

$$\begin{aligned} a + 2b + 2(-a - b) &= 0 \\ a + 2b - 2a - 2b &= 0 \\ -a &= 0 \\ a &= \frac{0}{-1} \\ a &= 0. \end{aligned}$$

$$\begin{aligned} c &= -0 - b \\ c &= -b. \end{aligned}$$

$$\begin{aligned} 0 + b - (-b) &= 0 \\ b + b &= 0 \\ 2b &= 0 \\ b &= \frac{0}{2} \\ b &= 0. \end{aligned}$$

$$\begin{aligned} c &= -0 \\ c &= 0. \end{aligned}$$

$$a = b = c = 0.$$

$$\begin{aligned} a(2, 1, 1) + b(2, 2, 1) + c(2, 2, -1) &= (x, y, z) \\ (2a, a, a) + (2b, 2b, b) + (2c, 2c, -c) &= (x, y, z) \\ (2a + 2b + 2c, a + 2b + 2c, a + b - c) &= (x, y, z). \end{aligned}$$

$$\begin{cases} 2a + 2b + 2c = x \\ a + 2b + 2c = y \\ a + b - c = z \end{cases}.$$

$$\begin{aligned} 2(a + b + c) &= x \\ a + b + c &= \frac{1}{2}x \end{aligned}$$

$$c = \frac{1}{2}x - a - b.$$

$$a + 2b + 2\left(\frac{1}{2}x - a - b\right) = y$$

$$a + 2b + x - 2a - 2b = y$$

$$-a + x = y$$

$$a = x - y.$$

$$c = \frac{1}{2}x - (x - y) - b$$

$$c = \frac{1}{2}x - x + y - b$$

$$c = \frac{-1}{2}x + y - b.$$

$$x - y + b - \left(\frac{-1}{2}x + y - b\right) = z$$

$$x - y + b + \frac{1}{2}x - y + b = z$$

$$\frac{3}{2}x - 2y + 2b = z$$

$$2b = \frac{-3}{2}x + 2y + z$$

$$b = \frac{\frac{-3}{2}x + 2y + z}{2}$$

$$b = \frac{-3}{4}x + y + \frac{1}{2}z$$

$$b = \frac{-3x + 4y + 2z}{4}.$$

$$c = \frac{-1}{2}x + y - \left(\frac{-3}{4}x + y + \frac{1}{2}z\right)$$

$$c = \frac{-1}{2}x + y + \frac{3}{4}x - y - \frac{1}{2}z$$

$$c = \frac{1}{4}x - \frac{1}{2}z$$

$$c = \frac{x - 2z}{4}.$$

Por lo tanto, este conjunto es base de $\mathbb{R}^3$.

(d) $\{(1, 2, 1), (1, 3, 1), (1, 4, 1), (1, 5, 1)\}.$

Este conjunto no es base de $\mathbb{R}^3$, ya que es linealmente dependiente.

(e) $\{(1, 1, 1), (-2, 1, 0), (-1, 0, 1)\}.$

$$a(1, 1, 1) + b(-2, 1, 0) + c(-1, 0, 1) = (0, 0, 0)$$

$$(a, a, a) + (-2b, b, 0) + (-c, 0, c) = (0, 0, 0)$$

$$(a - 2b - c, a + b, a + c) = (0, 0, 0).$$

$$\begin{cases} a - 2b - c = 0 \\ a + b = 0 \\ a + c = 0 \end{cases}.$$

$$b = -a.$$

$$c = -a.$$

$$a - 2(-a) - (-a) = 0$$

$$a + 2a + a = 0$$

$$4a = 0$$

$$a = \frac{0}{4}$$

$$a = 0.$$

$$b = -0$$

$$b = -0.$$

$$c = -0$$

$$c = 0.$$

$$a = b = c = 0.$$

$$a(1, 1, 1) + b(-2, 1, 0) + c(-1, 0, 1) = (x, y, z)$$

$$(a, a, a) + (-2b, b, 0) + (-c, 0, c) = (x, y, z)$$

$$(a - 2b - c, a + b, a + c) = (x, y, z).$$

$$\begin{cases} a - 2b - c = x \\ a + b = y \\ a + c = z \end{cases}.$$

$$b = y - a.$$

$$c = z - a.$$

$$a - 2(y - a) - (z - a) = x$$

$$a - 2y + 2a - z + a = x$$

$$4a - 2y - z = x$$

$$2a = x + 2y + z$$

$$a = \frac{x + 2y + z}{4}$$

$$a = \frac{1}{4}x + \frac{1}{2}y + \frac{1}{4}z$$

$$a = \frac{x + 2y + z}{4}.$$

$$b = y - \left(\frac{1}{4}x + \frac{1}{2}y + \frac{1}{4}z\right)$$

$$b = y - \frac{1}{4}x - \frac{1}{2}y - \frac{1}{4}z$$

$$b = \frac{-1}{4}x + \frac{1}{2}y - \frac{1}{4}z$$

$$b = \frac{-x + 2y - z}{4}.$$

$$c = z - \left(\frac{1}{4}x + \frac{1}{2}y + \frac{1}{4}z\right)$$

$$c = z - \frac{1}{4}x - \frac{1}{2}y - \frac{1}{4}z$$

$$c = \frac{-1}{4}x - \frac{1}{2}y + \frac{3}{4}z$$
$$c = \frac{-x-2y+3z}{4}.$$

Por lo tanto, este conjunto es base de $\mathbb{R}^3$.

Ejercicio 13 (*).

Sean $B = \{(2, 1, 1), (1, -1, 3), v\}$ una base de $\mathbb{R}^3$. Hallar v sabiendo que las coordenadas del vector $(1, -2, 5)$ en la base B son $(2, -1, 3)$.

Se sabe que:

$$B = \{(2, 1, 1), (1, -1, 3), v\}.$$

Entonces, se tiene:

$$[(1, -2, 5)]_B = (2, -1, 3).$$

Operando, se llega a:

$$(1, -2, 5) = 2(2, 1, 1) - (1, -1, 3) + 3v$$

$$(1, -2, 5) = (4, 2, 2) - (1, -1, 3) + 3v$$

$$(1, -2, 5) = (3, 3, -1) + 3v$$

$$3v = (1, -2, 5) - (3, 3, -1)$$

$$3v = (-2, -5, 6)$$

$$v = \frac{1}{3}(-2, -5, 6)$$

$$v = \left(\frac{-2}{3}, \frac{-5}{3}, 2\right).$$

Por lo tanto, $v = \left(\frac{-2}{3}, \frac{-5}{3}, 2\right)$.

Ejercicio 14.

Sean $B = \{(-1, 4, 2), v, (0, 0, -1)\}$ y $B' = \{w, (1, -1, 1), (-1, 0, 2)\}$ bases de $\mathbb{R}^3$. Hallar v y w sabiendo que las coordenadas de v en la base de B' son $(1, 2, 3)$ y que las coordenadas de w en la base B son $(1, 2, 3)$.

$$B = \{(-1, 4, 2), v, (0, 0, -1)\}.$$

$$B' = \{w, (1, -1, 1), (-1, 0, 2)\}.$$

$$[v]_{B'} = (1, 2, 3).$$

$$[w]_B = (1, 2, 3).$$

$$v = w + 2(1, -1, 1) + 3(-1, 0, 2)$$

$$v = w + (2, -2, 2) + (-3, 0, 6)$$

$$v = w + (-1, -2, 8).$$

$$w = (-1, 4, 2) + 2v + 3(0, 0, -1)$$

$$w = (-1, 4, 2) + 2v + (0, 0, -3)$$

$$w = 2v + (-1, 4, -1).$$

$$v = 2v + (-1, 4, -1) + (-1, -2, 8)$$

$$v = 2v + (-2, 2, 7)$$

$$2v - v = (-2, 2, 7)$$

$$v = (2, -2, -7).$$

$$w = 2(2, -2, -7) + (-1, 4, -1)$$

$$w = (4, -4, -14) + (-1, 4, -1)$$

$$w = (3, 0, -15).$$

Ejercicio 15.

Probar que los siguientes conjuntos son subespacios:

(a) $W = \{(x, y, z): x = y = z\}$ de $\mathbb{R}^3$.

W es cerrado para la suma:

$$\begin{aligned} (x_1, y_1, z_1), (x_2, y_2, z_2) \in W &\Rightarrow x_1 = y_1 = z_1 \wedge x_2 = y_2 = z_2 \\ &\Rightarrow x_1 + x_2 = y_1 + y_2 = z_1 + z_2 \\ &\Rightarrow (x_1 + x_2, y_1 + y_2, z_1 + z_2) \in W \\ &\Rightarrow (x_1, y_1, z_1) + (x_2, y_2, z_2) \in W. \end{aligned}$$

W es cerrado para la multiplicación de escalares:

$$\begin{aligned} (x, y, z) \in W &\Rightarrow x = y = z \\ &\Rightarrow \forall k \in \mathbb{R}, kx = ky = kz \\ &\Rightarrow (kx, ky, kz) \in W \\ &\Rightarrow k(x, y, z) \in W. \end{aligned}$$

Por lo tanto, el conjunto es un subespacio.

(b) $W = \{(x, y, z, t): x = z, y = t\}$ de $\mathbb{R}^4$.

W es cerrado para la suma:

$$\begin{aligned} (x_1, y_1, z_1, t_1), (x_2, y_2, z_2, t_2) \in W &\Rightarrow x_1 = z_1, y_1 = t_1 \wedge x_2 = z_2, y_2 = t_2 \\ &\Rightarrow x_1 + x_2 = z_1 + z_2, y_1 + y_2 = t_1 + t_2 \\ &\Rightarrow (x_1 + x_2, y_1 + y_2, z_1 + z_2, t_1 + t_2) \in W \\ &\Rightarrow (x_1, y_1, z_1, t_1) + (x_2, y_2, z_2, t_2) \in W. \end{aligned}$$

W es cerrado para la multiplicación de escalares:

$$\begin{aligned} (x, y, z, t) \in W &\Rightarrow x = z, y = t \\ &\Rightarrow \forall k \in \mathbb{R}, kx = kz, ky = kt \\ &\Rightarrow (kx, ky, kz, kt) \in W \\ &\Rightarrow k(x, y, z, t) \in W. \end{aligned}$$

Por lo tanto, el conjunto es un subespacio.

(c) $W = \{(x, y, z, t): 2y + 3z = 0\}$ de $\mathbb{R}^4$.

W es cerrado para la suma:

$$(x_1, y_1, z_1, t_1), (x_2, y_2, z_2, t_2) \in W \Rightarrow 2y_1 + 3z_1 = 0 \wedge 2y_2 + 3z_2 = 0$$

$$\begin{aligned}
&\Rightarrow 2y_1 + 2y_2 + 3z_1 + 3z_2 = 0 \\
&\Rightarrow 2(y_1 + y_2) + 3(z_1 + z_2) = 0 \\
&\Rightarrow (x_1 + x_2, y_1 + y_2, z_1 + z_2, t_1 + t_2) \in W \\
&\Rightarrow (x_1, y_1, z_1, t_1) + (x_2, y_2, z_2, t_2) \in W.
\end{aligned}$$

W es cerrado para la multiplicación de escalares:

$$\begin{aligned}
(x, y, z, t) \in W &\quad \Rightarrow 2y + 3z = 0 \\
&\Rightarrow \forall k \in \mathbb{R}, k(2y + 3z) = 0 \\
&\Rightarrow \forall k \in \mathbb{R}, 2ky + 3kz = 0 \\
&\Rightarrow (kx, ky, kz, kt) \in W \\
&\Rightarrow k(x, y, z, t) \in W.
\end{aligned}$$

Por lo tanto, el conjunto es un subespacio.

Ejercicio 16.

Una compañía petrolera puede convertir un barril de crudo en tres clases distintas de combustible. Sin aditivos de plomo, sus producciones de las tres clases de combustible a partir de un barril de crudo vienen dadas por el vector $(2, 2, 4)$. Con el máximo de aditivos de plomo permitidos legalmente, las producciones son $(5, 0, 3)$. Se supone que los efectos de los aditivos de plomo son proporcionales, es decir, que al usar una fracción del máximo permitido $(0 \leq \alpha \leq 1)$ se tiene la producción:

$$(1 - \alpha) (2, 2, 4) + \alpha (5, 0, 3).$$

Determinar si es posible que la compañía produzca los siguientes vectores:

(a) $(\frac{5}{2}, 1, \frac{7}{2})$.

$$(\frac{5}{2}, 1, \frac{7}{2}) = (1 - \alpha) (2, 2, 4) + \alpha (5, 0, 3)$$

$$(\frac{5}{2}, 1, \frac{7}{2}) = (2, 2, 4) + (-2\alpha, -2\alpha, -4\alpha) + (5\alpha, 0, 3\alpha)$$

$$(\frac{5}{2}, 1, \frac{7}{2}) = (2 + 3\alpha, 2 - 2\alpha, 4 - \alpha).$$

$$\begin{cases} \frac{5}{2} = 2 + 3\alpha \\ 1 = 2 - 2\alpha \\ \frac{7}{2} = 4 - \alpha \end{cases}$$

$$3\alpha = \frac{5}{2} - 2$$

$$3\alpha = \frac{1}{2}$$

$$\alpha = \frac{\frac{1}{2}}{3}$$

$$\alpha = \frac{1}{6}.$$

$$2\alpha = 2 - 1$$

$$2\alpha = 1$$

$$\alpha = \frac{1}{2}.$$

$$\alpha = 4 - \frac{7}{2}$$

$$\alpha = \frac{1}{2}.$$

Por lo tanto, no es posible que la compañía produzca este vector.

(b) $(\frac{9}{2}, \frac{1}{3}, \frac{19}{6})$.

$$(\frac{9}{2}, \frac{1}{3}, \frac{19}{6}) = (1 - \alpha) (2, 2, 4) + \alpha (5, 0, 3)$$

$$\begin{pmatrix} \frac{9}{2}, \frac{1}{3}, \frac{19}{6} \end{pmatrix} = (2, 2, 4) + (-2\alpha, -2\alpha, -4\alpha) + (5\alpha, 0, 3\alpha)$$

$$\begin{pmatrix} \frac{9}{2}, \frac{1}{3}, \frac{19}{6} \end{pmatrix} = (2 + 3\alpha, 2 - 2\alpha, 4 - \alpha).$$

$$\begin{cases} \frac{9}{2} = 2 + 3\alpha \\ \frac{1}{3} = 2 - 2\alpha. \\ \frac{19}{6} = 4 - \alpha \end{cases}$$

$$3\alpha = \frac{9}{2} - 2$$

$$3\alpha = \frac{5}{2}$$

$$\alpha = \frac{\frac{5}{2}}{3}$$

$$\alpha = \frac{5}{6}.$$

$$2\alpha = 2 - \frac{1}{3}$$

$$2\alpha = \frac{5}{3}$$

$$\alpha = \frac{\frac{5}{3}}{2}$$

$$\alpha = \frac{5}{6}.$$

$$\alpha = 4 - \frac{19}{6}$$

$$\alpha = \frac{5}{6}.$$

Por lo tanto, sí es posible que la compañía produzca este vector.

(c) $(1, 6, 9)$.

$$(1, 6, 9) = (1 - \alpha) (2, 2, 4) + \alpha (5, 0, 3)$$

$$(1, 6, 9) = (2, 2, 4) + (-2\alpha, -2\alpha, -4\alpha) + (5\alpha, 0, 3\alpha)$$

$$(1, 6, 9) = (2 + 3\alpha, 2 - 2\alpha, 4 - \alpha).$$

$$\begin{cases} 1 = 2 + 3\alpha \\ 6 = 2 - 2\alpha. \\ 9 = 4 - \alpha \end{cases}$$

$$3\alpha = 1 - 2$$

$$3\alpha = -1$$

$$\alpha = \frac{-1}{3}.$$

$$2\alpha = 6 - \frac{1}{3}$$

$$2\alpha = \frac{17}{3}$$

$$\alpha = \frac{\frac{17}{3}}{\frac{2}{6}}$$

$$\alpha = \frac{17}{6}.$$

$$\alpha = 4 - 9$$

$$\alpha = -5.$$

Por lo tanto, no es posible que la compañía produzca este vector.

Ejercicio 17.

Considerar el conjunto:

$$V = \{(w, td, ti, P, GP) : P = w, td + ti = GP\},$$

donde w es el crecimiento de los salarios nominales, td es el crecimiento de los impuestos directos, ti es el crecimiento de los impuestos indirectos, P es el crecimiento de los precios y GP es el crecimiento del gasto público.

(a) Mostrar que V es un subespacio vectorial de $\mathbb{R}^5$.

V es cerrado para la suma:

$$\begin{aligned} (w_1, td_1, ti_1, P_1, GP_1), (w_2, td_2, ti_2, P_2, GP_2) \in V &\Rightarrow P_1 = w_1, td_1 + ti_1 = GP_1 \wedge P_2 = w_2, \\ td_2 + ti_2 = GP_2 & \\ \Rightarrow P_1 + P_2 = w_1 + w_2, (td_1 + td_2) + (ti_1 + ti_2) = GP_1 + GP_2 & \\ \Rightarrow (w_1 + w_2, td_1 + td_2, ti_1 + ti_2, P_1 + P_2, GP_1 + GP_2) \in V & \\ \Rightarrow (w_1, td_1, ti_1, P_1, GP_1) + (w_2, td_2, ti_2, P_2, GP_2) \in V. & \end{aligned}$$

V es cerrado para la multiplicación de escalares:

$$\begin{aligned} (w, td, ti, P, GP) \in V &\Rightarrow P = w, td + ti = GP \\ \Rightarrow \forall k \in \mathbb{R}, kP = kw, k(td + ti) = kGP & \\ \Rightarrow \forall k \in \mathbb{R}, kP = kw, ktd + kti = kGP & \\ \Rightarrow (kw, ktd, kti, kP, kGP) \in V & \\ \Rightarrow k(w, td, ti, P, GP) \in V. & \end{aligned}$$

Por lo tanto, V es un subespacio vectorial de $\mathbb{R}^5$.

(b) Hallar una base y la dimensión de V . Dar una interpretación económica del resultado.

$$(P, td, ti, P, td + ti) = P(1, 0, 0, 1, 0) + td(0, 1, 0, 0, 1) + ti(0, 0, 1, 0, 1).$$

$$B_V = \{(1, 0, 0, 1, 0), (0, 1, 0, 0, 1), (0, 0, 1, 0, 1)\}.$$

$$\dim(V) = 3.$$